

Tutorial 5_summary

Tutorial 5 — Reliability, TLOC, Capacitive/Inductive/Active Sensing & Mechanics

Source: Tutorial 5 with Solutions.pdf
Instructor: Qilong Zhong, MSc, GTIIT
Corresponds to: Lectures 5–6 (Chapters 7–8 of Bentley)

1. Reliability Assessment

1.1 Core Definitions

Metric	Symbol	Formula	Meaning
Reliability	$R(t)$	—	Probability of normal operation within specified environment & time
Unreliability	$F(t)$	$F = 1 - R$	Probability of failure
MTTF	—	$\sum T_i / N$	Average lifespan (non-repairable)
MDT	—	Total Down Time / Number of Failures	Average repair duration
MTBF	—	Total Up Time / Number of Failures	Time between two failures
Failure rate	$\bar{\lambda}$	$1/\text{MTBF}$	Failures per unit time
Availability	A	$\frac{\text{MTBF}}{\text{MTBF} + \text{MDT}}$	Probability system is operational

✔ **Golden rule:** $R(t) + F(t) = 1$ — equipment is either working or not.

2. System Architecture Reliability

2.1 Constant Failure Rate Model

For components in useful life: $\lambda = \text{constant}$.

$$R(t) = e^{-\lambda t}$$

$$F(t) = 1 - e^{-\lambda t}$$

2.2 Series Systems

Logic: ALL components must work for the system to operate.

Rule	Formula
Multiply reliabilities	$R_{SYS} = R_1 \times R_2 \times \dots \times R_n$
Add failure rates	$\lambda_{SYS} = \lambda_1 + \lambda_2 + \dots + \lambda_n$

▲ Design principle: "Do not multiply entities beyond necessity" — minimize component count.

2.3 Parallel Systems

Logic: System works as long as at least ONE component works (fails only if ALL fail).

Rule	Formula
Multiply unreliabilities	$F_{SYS} = F_1 \times F_2 \times \dots \times F_n$
Identical components	$F_{SYS} = F^n$

✓ Parallelism is a "heavy hitter" for improving reliability (probabilities shrink when multiplied), but at the cost of doubled hardware.

2.4 Mixed Redundancy (Median Selector)

Principle: Three sensors in parallel → output via **median selector** → connected in series with other components.

Example (3 thermocouples, $\lambda_1 = 1.1$, + converter + recorder, each $\lambda = 0.1$, $t = 0.5$ yr):

Step	Calculation	Result
Single TC unreliability	$F = 1 - e^{-1.1 \times 0.5}$	$1 - e^{-0.55}$
3-way parallel unreliability	$F_1 = (1 - e^{-0.55})^3$	0.076
3-way parallel reliability	R_1	0.924
Series components	$R_2 = R_3 = R_4 = e^{-0.1 \times 0.5}$	0.951 each
Overall system	$R = 0.924 \times 0.951^3$	0.795

3. Total Life Cycle Operating Cost (TLOC)

3.1 Definition

For any given application, the best system is the one with the **minimum TLOC**.

$$TLOC = \text{Initial Purchase Cost} + \text{Total Failure/Maintenance Cost} + \text{Total Measurement Error Cost}$$

3.2 Simplified Comparison Formula

When measurement errors are identical and there's no preventive maintenance:

$$TLOC = C_1 + [C_R + (C_L + C_P) \times T_R] \times \lambda \times T$$

Term	Meaning
C_1	Initial purchase cost
C_R	Material cost per repair
C_L	Labor cost per hour
C_P	Process downtime cost per hour
T_R	Repair time (hours)
λ	Annual failure rate
T	Total lifespan (years)

▲ C_P is the hidden giant: Process downtime loss is often **hundreds or thousands of times** the spare parts cost. "Repairing quickly" and "not breaking easily" are extremely valuable.

3.3 Worked Example (10-year lifespan)

	System 1	System 2
Initial cost	£1,000	£2,000
Cost per failure	£900	£1,335
Failures over 10 years	20	10
Total maintenance	£18,000	£13,350
TLOC	£19,000	£15,350 ✓

✓ System 2 costs **twice as much** initially but has **lower TLOC** because of lower failure rate.

4. Capacitive Sensing Components

4.1 Fundamental Formula

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

C depends on: ϵ_0 (vacuum permittivity), ϵ_r (relative permittivity), A (plate overlap area), d (plate separation).

4.2 Three Types

Type	What Changes	Application
Variable distance	d	Pressure sensors (diaphragm deflection)
Variable area	A	Linear displacement
Variable dielectric	ϵ_r	Liquid level, material boundary detection

4.3 Variable Dielectric Sensor — Equivalent Circuit

When a dielectric plate is inserted between the plates:

$$C_{total} = C_{air} + C_{dielectric} = \frac{\epsilon_0 w}{d} [\epsilon_1 x + \epsilon_2 (l - x)]$$

✔ **Perfectly linear** relationship between C and displacement x .

5. Variable Reluctance Sensing Components

5.1 Magnetic Circuit Fundamentals

Quantity	Formula
Ohm's Law for magnetic circuits	$\mathcal{F} = \Phi \times \mathcal{R}$
Inductance-Reluctance relation	$L = \frac{N^2}{\mathcal{R}_{total}}$
Reluctance	$\mathcal{R} = \frac{l}{\mu \mu_0 A}$

5.2 Variable Reluctance Displacement Sensor

$$\mathcal{R}_{total} = \mathcal{R}_{core} + \mathcal{R}_{armature} + \mathcal{R}_{gap}$$

Material	μ	Reluctance
Ferromagnetic (iron)	~100–1000	Very low
Air	1	Extremely high

▲ **Critical insight:** Even if the air gap d is only 1 mm, its reluctance may be **far greater** than a thick iron core! A tiny increase in $d \rightarrow$ total reluctance rises sharply $\rightarrow$ inductance L plunges non-linearly.

6. Active Electrical Sensing Elements

6.1 What Does "Active" Mean?

Self-generating: Does NOT require external power supply. Extracts energy directly from the measurand (e.g., mechanical kinetic energy) and converts it into an AC electrical signal.

6.2 Faraday's Law of Induction

$$E = -\frac{d\Psi}{dt}$$

Induced EMF = negative rate of change of magnetic flux linkage.

6.3 Mathematical Model — Variable Reluctance Tachometer

- Flux linkage varies sinusoidally with angular displacement: $\Psi(\theta) \approx a + b \cos(m\theta)$
- m = number of gear teeth
- ω_r = angular velocity of gear ($d\theta/dt$)

$$E = bm\omega_r \sin(m\omega_r t)$$

6.4 Two Direct Proportionality Laws

Law	Formula	Meaning
Amplitude	$\hat{E} = bm\omega_r$	Peak voltage $\propto$ rotational speed
Frequency	$f = \frac{m\omega_r}{2\pi} = \frac{RPM}{60} \times m$	AC signal frequency $\propto$ rotational speed

7. Mechanical Mechanics & Elastic Deformation

7.1 Stress and Strain

Quantity	Definition	Formula
Stress σ	Internal force per unit area	$\sigma = F/A$ (+ tensile, – compressive)
Strain e	Relative change in dimension	$e = \Delta l/l$
Young's Modulus E	Stiffness of material	$E = \sigma/e$

7.2 Poisson's Effect (3D Deformation Linkage)

Physical phenomenon: When stretched (lengthened), material **must** become thinner. When compressed (shortened), it **must** become thicker.

$$e_T = -\nu e_L$$

Symbol	Meaning	Typical Value
e_L	Longitudinal strain	—
e_T	Transverse strain	—
ν	Poisson's ratio	0.25–0.4

▲ The **negative sign** is vital! Transverse and longitudinal deformation are always in **opposite** directions. Compressive force $\rightarrow e_L < 0 \rightarrow e_T > 0$ (material bulges/thickens).

8. Dynamic Characteristics of Sensors

8.1 Three Elements of a Real Sensor

Element	Role
Mass / Inertia (m, I)	Resists sudden changes in motion
Damping / Friction (λ, b)	Dissipates energy, hinders relative motion
Stiffness / Elasticity (k, c)	Provides restorative force during deformation

8.2 Three Core Dynamic Parameters

Parameter	Formula (Translation)	Formula (Rotation)
Steady-state gain K	—	—
Natural frequency ω_n	$\sqrt{k/m}$	$\sqrt{c/I}$
Damping ratio ξ	$\lambda/(2\sqrt{km})$	—

8.3 Accelerometer Principle — Relative Displacement

The outer casing vibrates with the measured object. The internal proof mass **lags behind** due to inertia $\rightarrow$ **relative displacement** between mass and casing is what the sensor measures.

8.4 Amplitude-Frequency Characteristic

$$\frac{\hat{\theta}}{\hat{\phi}_i} = \frac{(\omega/\omega_n)^2}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\xi(\omega/\omega_n)]^2}}$$

8.5 Engineering's Golden Damping Ratio

$$\xi = 0.707 = \frac{1}{\sqrt{2}}$$

✔ This ratio enables the **flattest possible response** over the widest frequency range. Mathematically: the complex denominator simplifies to $\sqrt{1 + (\omega/\omega_n)^4}$, greatly reducing calculation difficulty.

9. Summary Table

Domain	Key Concept	Core Formula
Reliability	Exponential model	$R(t) = e^{-\lambda t}$
Series system	Add failure rates	$\lambda_{SYS} = \sum \lambda_i$
Parallel system	Multiply unreliabilities	$F_{SYS} = \prod F_i$
TLOC	Total ownership cost	$C_1 + [C_R + (C_L + C_P)T_R]\lambda T$
Capacitive	Variable dielectric	$C_{total} = \frac{\epsilon_0 w}{d} [\epsilon_1 x + \epsilon_2 (l - x)]$
Inductive	Variable reluctance	$L = N^2 / \mathcal{R}_{total}$
Active (EM)	Faraday's Law	$E = -d\Psi/dt$
Strain/Poisson	3D deformation	$e_T = -\nu e_L$
Accelerometer	Relative displacement	$\frac{\hat{\theta}}{\hat{\phi}_i} = f(\omega/\omega_n, \xi)$

✔ **Key Takeaway:** Reliability is quantified by λ and modeled as exponential decay $R = e^{-\lambda t}$. Series systems multiply risk ($\lambda_{SYS} = \sum \lambda_i$); parallel redundancy divides it ($F_{SYS} = \prod F_i$). TLOC reveals that higher initial cost often means lower lifetime cost. Capacitive, inductive, and electromagnetic sensing each exploit a different physical domain. Accelerometer design targets $\xi = 0.707$ for the flattest frequency response.